

Excel Formulas Handwritten Notes



Complete Formula Guide - Page 1

★ 1. Formula Categories

- Basic Math
- Logical
- Conditional
- Lookup
- Text
- Date & Time
- Dynamic Array
- Advanced

Know the type
of formula before
you use it!

2. Basic Syntax

- Formula starts with $\equiv$
- Cell reference examples: A1, B2:C10
- Use comma (,) as separator
- Use parentheses () for order

	A	B	C
1	A1		
2		B2:C10	
3			

3. Operators

- + : addition
- : subtraction
- * : multiplication
- / : division
- ^ : power
- & : join text

Comparison Operators

- = : equal to
- > : greater than
- < : less than
- >= : greater than or equal to
- <= : less than or equal to
- <> : not equal to



Tip: Learn formulas category by category
and practice on small datasets.



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★ Basic Math & Stats Formulas ✧ ✧

1. SUM

→ Syntax :

`= SUM(A1:A10)`

→ Use :

Adds all the values in the range.

Example :

`= SUM(A1:A10)`

→ returns total



2. AVERAGE

→ Syntax :

`= AVERAGE(B1:B10)`

→ Use :

Returns the mean (average) of values.

Example :

`= AVERAGE(B1:B10)`

→ returns average



3. MIN and MAX

→ Syntax :

- `= MIN(C1:C10)`
- `= MAX(C1:C10)`

→ Use :

Returns the smallest value (MIN) or largest value (MAX).

Example :

- `MIN(C1:C10)` → smallest
- `MAX(C1:C10)` → largest



4. COUNT and COUNTA

→ Syntax :

- `= COUNT(A1:A10)`
- `= COUNTA(A1:A10)`

Use :

- COUNT → counts numbers only
- COUNTA → counts non-empty cells (texts, numbers, etc.)

Example :

`= COUNT(A1:A10)` → numbers

`= COUNTA(A1:A10)` → non-blank cells



5. ROUND, ROUNDUP, ROUNDDOWN

→ Syntax :

- `= ROUND(D2, 2)`
- `= ROUNDUP(D2, 2)`
- `= ROUNDDOWN(D2, 2)`

Use :

Rounds a number to given decimal places.

Example :

- `= ROUND(D2, 2)` → 123.46
- `= ROUNDUP(D2, 2)` → 123.46
- `= ROUNDDOWN(D2, 2)` → 123.45

123.456

6. ABS and SQRT

→ Syntax :

- `= ABS(-25)`
- `= SQRT(144)`

Use :

- ABS → returns absolute value (positive value)
- SQRT → returns square root of a number

Example :

`= ABS(-25)` → 25

`= SQRT(144)` → 12

$|-25|$
 $= 25$

$\sqrt{144}$
 $= 12$

★ Note: COUNT counts numbers only. COUNTA counts non-blank cells.

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Logical & Conditional Formulas

1. IF

→ =IF(A2>=40,"Pass","Fail")

- Returns one value if TRUE, another if FALSE.

A2 >= 40 ✓ Pass

A2 < 40 ✗ Fail

2. AND

→ =AND(A2>50, B2="Yes")

- TRUE if all conditions are TRUE.

A2 > 50
B2 = "Yes" → TRUE ✓

3. OR

→ =OR(C2="Paid", D2="Pending")

- TRUE if any condition is TRUE.

C2 = "Paid"
D2 = "Pending" → TRUE ✓

4. NOT

→ =NOT(E2="Closed")

- Reverses the logical value.

E2 = "Closed" → FALSE ✗
↓
NOT(...) → TRUE ✓

5. IFS

→ =IFS(A2>=90,"A", A2>=75,"B", A2>=60,"C")

- Checks multiple conditions in order.

>= 90 → "A"
>= 75 → "B"
>= 60 → "C"

6. IFERROR

→ =IFERROR(F2/G2,"Error")

- Returns value or custom message if error.

F2/G2 → ✓ or ✗
↓ ↓
Value "Error"

7. SUMIF

→ =SUMIF(A:A,"North", B:B)

- Adds values in B where A matches "North".

Region	Sales
North	120
North	80
South	60

→ 200

8. COUNTIF

→ =COUNTIF(A:A,"Active")

- Counts cells that meet the criteria.

✓ ✓ ✗ ✓ ✓ ✗ ✓ → 5

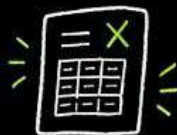
9. AVERAGEIF

→ =AVERAGEIF(A:A,"North", C:C)

- Average of values in C where A is "North".

North → 50, 70, 90 → 70

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★ Lookup & Reference Formulas

1. **VLOOKUP** → `=VLOOKUP(A2, $F$2:$H$10, 2, FALSE)`

Looks for A2 in first column of table and returns value from 2nd column.

	F	G	H
2	A2	Return Cell	
3			
...			
10			

2. **HLOOKUP** → `=HLOOKUP(B1, $A$1:$G$5, 3, FALSE)`

Looks for B1 in first row of table and returns value from 3rd row.

	A	B	...	G
1				
2				
3		Return Cell		
5				

3. **XLOOKUP** → `=XLOOKUP(A2, F:F, G:G, "Not Found")`

Looks for A2 in column F and returns matching value from column G. If not found, returns "Not Found".

	F	G
	A2	
	...	Return Cell

4. **INDEX** → `=INDEX(C2:C10, 3)`

Returns the 3rd value from C2:C10.

C
C2
C3
C4
...
C10

5. **MATCH** → `=MATCH("Apple", A2:A10, 0)`

Returns position of "Apple" in A2:A10. Exact match (0).

A
Apple
Banana
Cherry
...
Grape

← Returns position = 1

6. **INDEX + MATCH** → `=INDEX(B2:B10, MATCH(E2, A2:A10, 0))`

Finds E2 in A2:A10, gets position, then returns value from same row in B2:B10.

A	B
Item1	Value1
Item2	Value2
Item3	Value3
...	...

E2
Item2

7. **OFFSET** → `=OFFSET(A1, 2, 1, 3, 2)`

Returns a range that is 2 rows down, 1 column right from A1, with height 3 and width 2.

	A	B	C	D	E
1					
2					
3					
4					
5					

↓ 2 rows

→ 1 col

★ **VLOOKUP**: older but common

★ **XLOOKUP**: more flexible



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★ Text Formulas

1. **LEFT** → **=LEFT(A2,3)**
 - Returns the first N characters from a text string.
2. **RIGHT** → **=RIGHT(A2,4)**
 - Returns the last N characters from a text string.
3. **MID** → **=MID(A2,2,5)**
 - Returns a specific number of characters from the middle of a text string.
4. **LEN** → **=LEN(A2)**
 - Returns the total number of characters in a text string.
5. **TRIM** → **=TRIM(A2)**
 - Removes extra spaces from text (leaving single spaces between words).
6. **CONCAT** and **CONCATENATE** → combine text
 - **=CONCAT(A2," ",B2)** or **=CONCATENATE(A2," ",B2)**
7. **TEXTJOIN** → **=TEXTJOIN("","",TRUE,A2:A5)**
 - Joins text from a range with a delimiter. TRUE ignores empty cells.
8. **UPPER, LOWER, PROPER**
 - **=UPPER(A2)** → all uppercase
 - **=LOWER(A2)** → all lowercase
 - **=PROPER(A2)** → capitalize each word
9. **FIND** and **SEARCH** → find text within text
 - **=FIND("a",A2)** → case-sensitive
 - **=SEARCH("a",A2)** → case-insensitive
10. **SUBSTITUTE** → **=SUBSTITUTE(A2,"-",",")**
 - Replaces old text with new text.

“Text formulas help you extract, clean, and combine text like a pro!”

FIND is case-sensitive,
SEARCH is not.

“Clean text, better data, smarter sheets!”

Tip:



Use these formulas to clean names, format IDs, and build better reports faster!



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★ Date & Time Formulas



1. TODAY → =TODAY() - Returns today's date.
2. NOW → =NOW() - Returns current date & time.
3. DATE → =DATE(2026,6,23) - Creates a date from year, month, day.
4. DAY, MONTH, YEAR }
=DAY(A2)
=MONTH(A2)
=YEAR(A2) } Extract date parts from a date.
5. DATEDIF → =DATEDIF(A2,B2,"y") - Difference in years between two dates.
6. EDATE → =EDATE(A2,3) - Returns date 3 months after the date in A2.
7. EOMONTH → =EOMONTH(A2,0) - Returns last day of the month for the date in A2.
8. NETWORKDAYS → =NETWORKDAYS(A2,B2) - Returns number of working days between two dates.
9. TEXT → =TEXT(A2,"dd-mmm-yyyy")
- Formats a date as text (example: 23-Jun-2026).



Remember:

Excel stores dates as serial numbers.



JUN 2026						
M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

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Dynamic Array & Filter Formulas

1. **FILTER** → `=FILTER(A2:C10, C2:C10 = "Yes")`
Filters rows that meet the given condition.
2. **SORT** → `=SORT(A2:B10, 2, -1)`
Sorts by the 2nd column in descending order (-1).
3. **SORTBY** → `=SORTBY(A2:B10, B2:B10, -1)`
Sorts by range B2:B10 in descending order.
4. **UNIQUE** → `=UNIQUE(A2:A20)`
Returns a list of unique values.
5. **SEQUENCE** → `=SEQUENCE(5, 1, 1, 1)`
Generates a sequence (rows, columns, start, step).
Example: 5 rows, 1 column, start at 1, step 1.
6. **TRANSPOSE** → `=TRANSPOSE(A1:C3)`
Converts rows to columns (or vice versa).
7. **RAND**
RANDBETWEEN →
 - `=RAND()` → random decimal between 0 and 1
 - `=RANDBETWEEN(1, 100)` → random whole number between 1 and 100
8. **SMALL**
LARGE →
 - `=SMALL(A2:A20, 3)` → returns 3rd smallest value
 - `=LARGE(A2:A20, 2)` → returns 2nd largest value

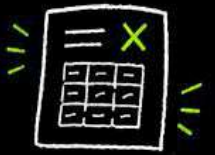
Note:



Dynamic array formulas can spill into nearby cells.

	A	B	C	D	E
1	=FILTER(...)	→	→	→	→
2		↓	↓	↓	↓
3					

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Advanced Formulas + Quick Practice Tips



1. SUMPRODUCT → = SUMPRODUCT (A2:A5, B2:B5)

2. SUBTOTAL → = SUBTOTAL (9, B2:B20)

3. INDIRECT → = INDIRECT ("A"&B2)

4. INDEX-MATCH recap → Use INDEX to return value, MATCH to find position.

5. TEXT function → = TEXT (A2, "0.00")

6. IF + AND combined example →
= IF (AND (A2 >= 50, B2 = "Yes"), "Selected", "No")

7. Nested formula reminder → keep brackets balanced ()

Practice Tips

- Practice on sales data
- Practice on salary sheet
- Practice on attendance
- Practice on marksheet
- Practice on dashboard tasks



Best way to learn Excel = understand + practice + repeat